



NORTHERN

Northern TLB510 Wheel Balancer User Manual

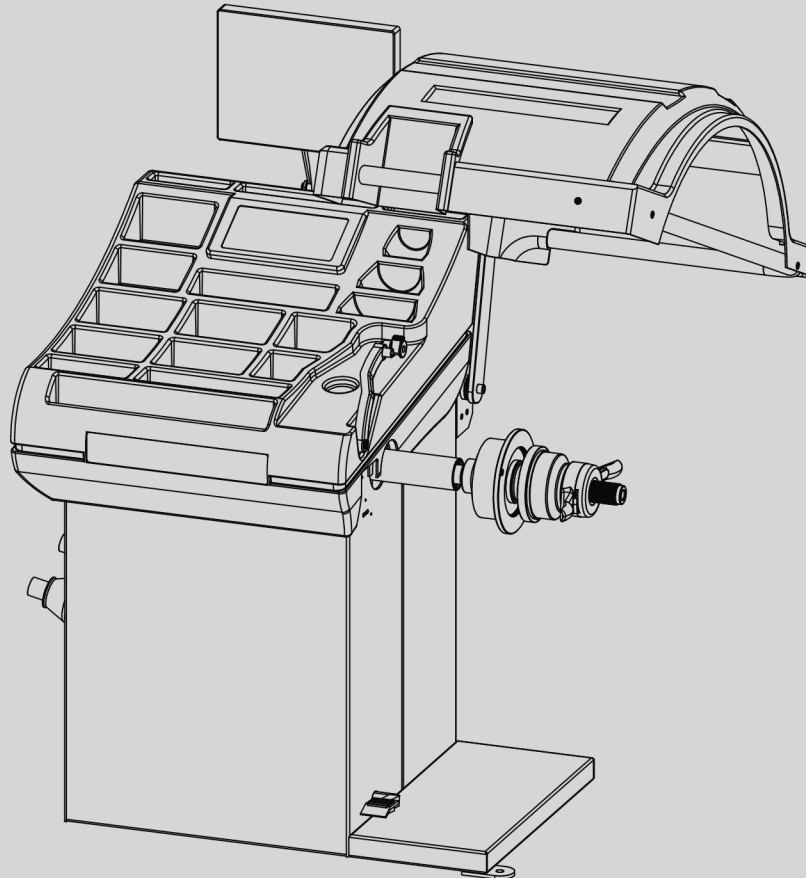


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Rated voltage	220V / 50Hz
Hub diameter	10" - 26"
Hub width	14" - 36"
Balancing precision	1g
Balancing time	7S (20Kg wheel)
Maxium tire weight	65Kg
Motor power	250W
Working noise	70db
Carton size	970*770*1150mm
Gross weight	125KG

Please check the products immediately after unpacking to ensure that they are in good condition.

Chapter I Safety Precautions

- Incorrect operation may cause personal injury and equipment damage.
- Please read carefully and understand all the contents of the instructions before use.
- Please ensure that children and other unauthorized personnel stay away from the work area.
- Ensure that the equipment is connected to the correct power and air source, and is reliably grounded.
- Please use this equipment on a flat, level, dry and reliable bearing plane.
- Avoid accidental startup. Please ensure that the equipment is turned off and the electrical source is disconnected before maintenance.
- Keep the protection device and safety device in the correct position and keep working normally.
- Keep the work area clean and well lit. Chaos or dark areas may cause accidents.
- It is strictly prohibited to overload this product, otherwise the resulting accident liability will not be covered by the insurance.
- Please keep away from heat and fire source. High temperature may cause damage to this equipment and sealing elements.
- Avoid dangerous environment. Do not use equipment in humid environment or expose it to rain.
- It is strictly prohibited for any untrained personnel to use this equipment, and it is not allowed to disassemble or refit this equipment voluntarily.
- Ensure that the wheels are installed correctly, and correct way is selected according to different wheel hubs to lock and fix them on the equipment.
- Check carefully before each use. If there is oil leakage, loosen or damaged parts and accessories, they cannot be used.
- Please let professionals with professional maintenance qualifications maintain the equipment reasonably. If accessory replacement is required, please use original accessory.
- Safety shoes, safety goggles and working gloves conforming to relevant national safety regulations must be worn during operation.
- It is strictly prohibited to use the equipment after drinking, mental fatigue, inattention, drowsiness and any unconsciousness caused by drugs.
- Before balancing, the tire and rim shall be checked for potential defects; it is prohibited to balance any defective tire and rim.
- The tire weight shall not exceed the bearing capacity of the balancing machine as overweight of the wheel may damage the balancing machine permanently; the balancing machine shall not be used for tires larger than design dimensions.

Warning

The notices, warnings, instructions and other information contained in the instructions cannot cover all possible situations. Operators must understand that daily prudent operation and professional knowledge are indispensable factors in operating this product.



1.1 Definition of risk level

Risk level referred to in the Manual applies to the following definitions and signs. Danger: Refers to direct dangers, which may cause serious personal injury or death.



Warning: Refers to danger or unsafe behavior, which may cause serious personal injury or death.

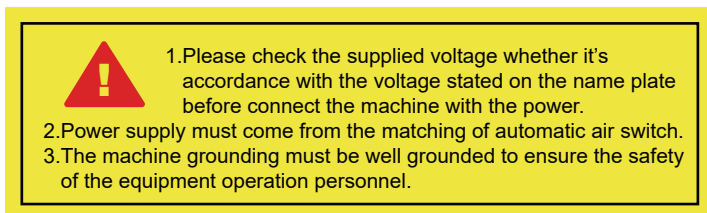


Notice: Refers to danger or unsafety, which may cause minor personal injury or product or property loss.



Caution: Watch out ! Which may endanger your safety or others

1.2 Sign and nameplate



Warning sign for power on



Sign for power supply specifications



Sign for rotation of transmission components

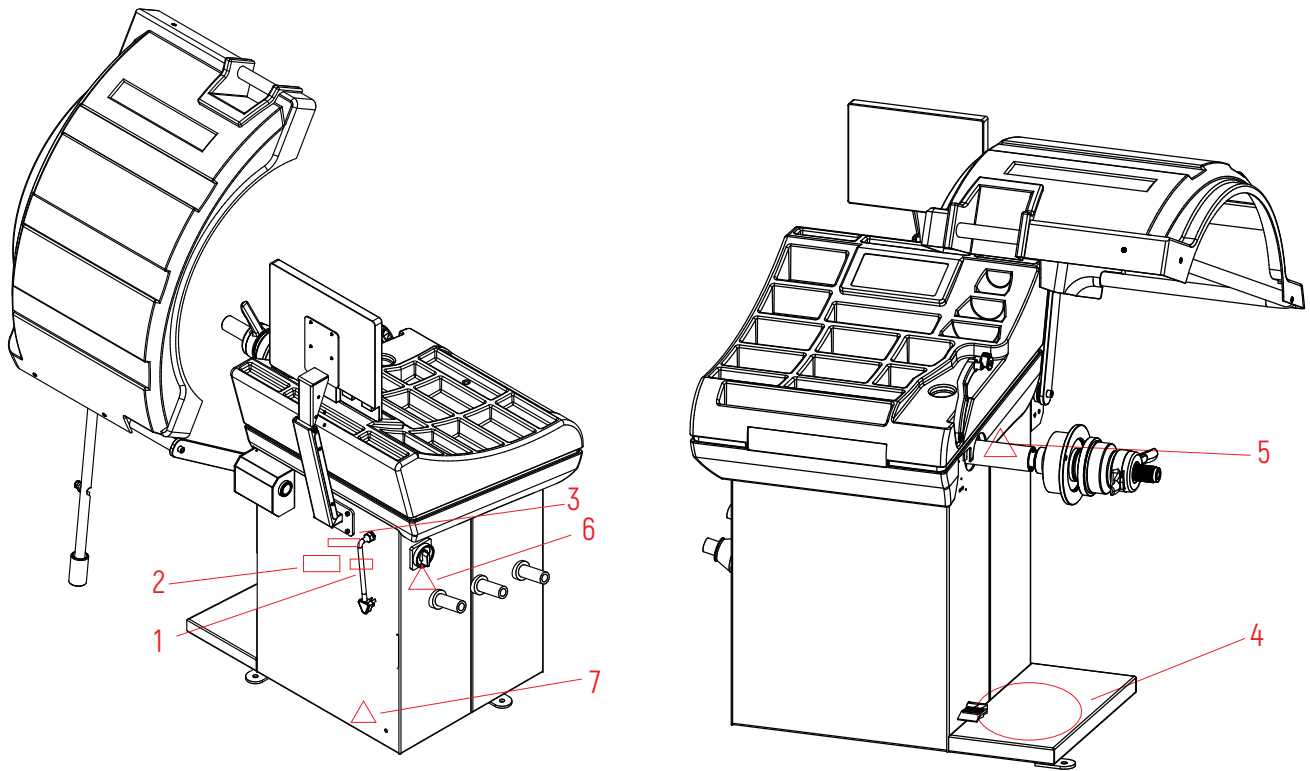


Sign for electric shock



Sign for grounding power supply

1.3 Position of sign and nameplate



Note: Please check and confirm whether the signs and nameplates are clear and complete according to the diagram above, and check whether the corresponding position is correct. In case of missing or incorrect position, please contact the local dealer.

1.4 Standard safety equipment

The equipment can stop the rotation of main rotating axle and tire under emergency.

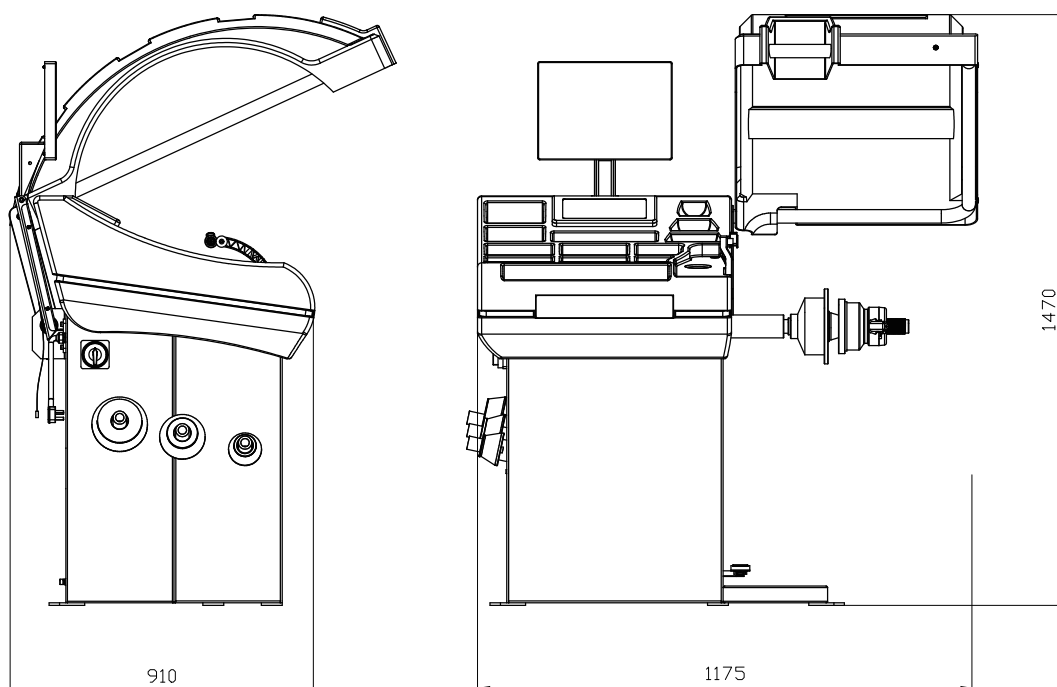


Chapter II Installation Instruction

The installation of tire balancing machine must be completed by professionals. Safe and effective use depends on correct installation. If you have any questions, please contact the authorized dealer.

2.1 Equipment size and use space

- The tire balancing machine must be placed on a firm flat floor and fixed with bolts.
- The installation position of the tire balancing machine must be kept away from the damp environment, and must be close to the power supply.
- Enough space must be reserved behind the selected position to ensure normal working of protective cover.
- At least 500 mm operating space must be reserved on the right and front of the tire balancing machine for tire dismounting and testing.

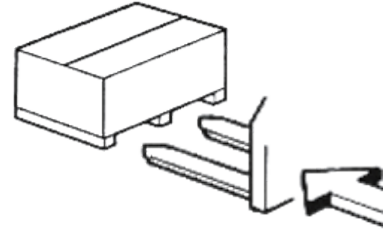


2.2 Safety rules

- This equipment shall be operated by professionals or trained personnel.
- The company will not be responsible for unauthorized modification of equipment (especially for electrical parts).
- Any treatment for electrical parts can only be carried out by professionals.

2.3 Transportation/unpacking

- Forklift handling is adopted, and moving position is as shown in the right figure.
- Unpack and check whether the equipment is damaged.
- Keep packing materials away from children to avoid danger.



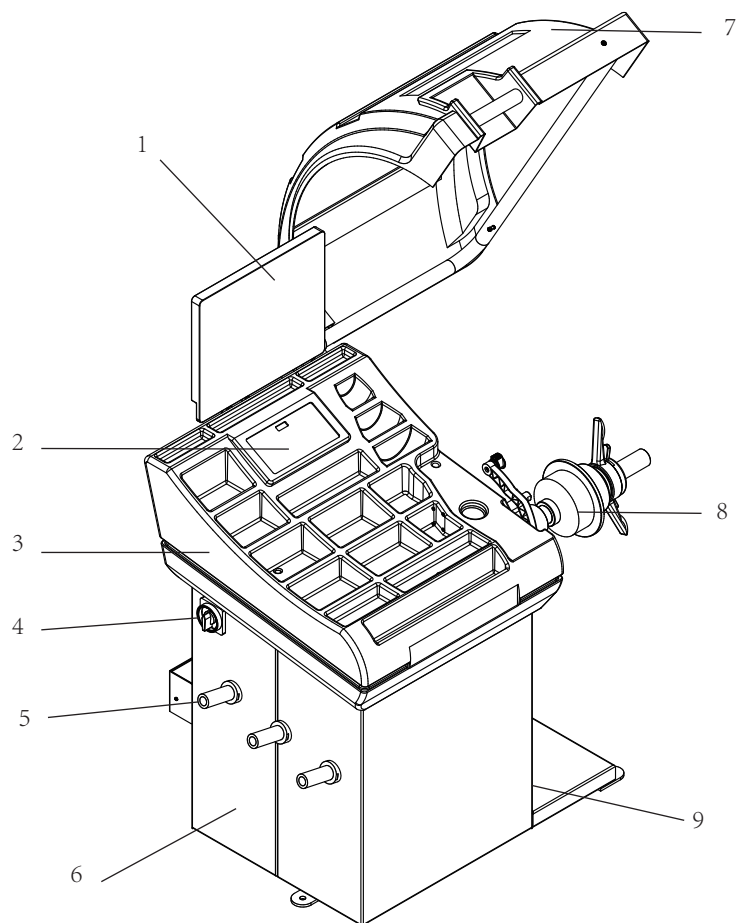
Note

The surface of equipment is coated with a layer of special anti-rust oil, which is easy to be coated with dust and shall be wiped off as much as possible when necessary.

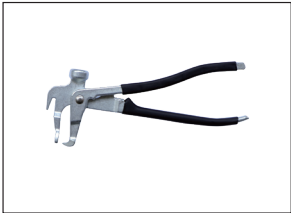


2.4 Product diagram

1. Display
2. Control panel
3. Upper plastic cover
4. Power switch
5. Hanging handle
6. Power cord
7. Protective cover
8. Drive subassembly
9. Wheel stop pedal



2.5 Standard accessories



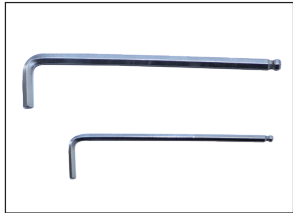
Balance plier



Self-calibration lead (100g)



Instructions



5 mm Allen wrench 8 mm
Allen wrench



Caliper



100-132mm



74-109mm



54-79mm



44-65mm



Half nuts



Screw rod of main axle



Screw



Rubber washer of half nut



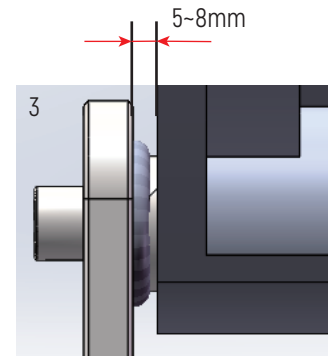
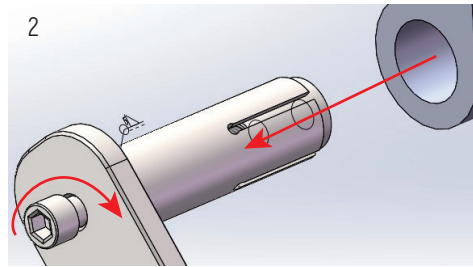
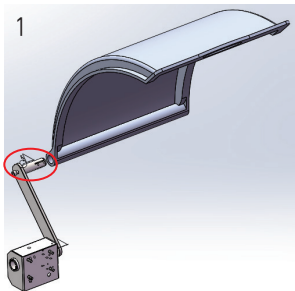
Pressure ring of half nut



Plastic bowl

2.6 Mounting of protective cover

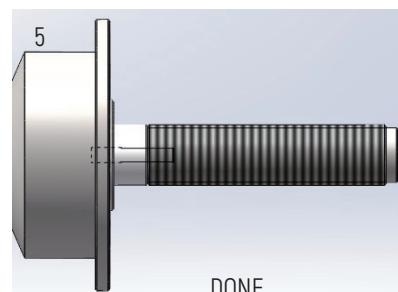
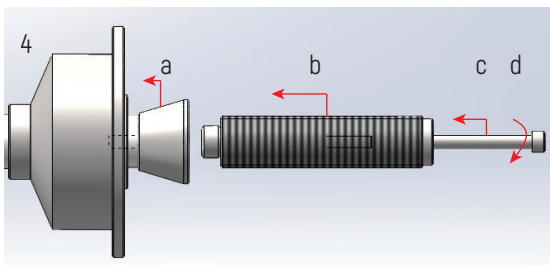
Instructions on protective cover installation 1. Keep the adjusting end of protective cover link mechanism perpendicular to the tank, and form an angle of 60° on the other end. Lock the screw tightly to avoid loosening. 2. Take out the protective cover and adjust it to the position 460 mm away from the screw rod as shown in the diagram, and then lock the screw tightly to avoid loosening. 3. Complete the installation, and keep it opening.



DONE

2.7 For installation of screw rod

Install the screw before it must first be two matched on the surface of the oil and other foreign bodies clean, so as not to affect the accuracy of measurement. The lead screw shall be fixed onto the balance axle with a spanner and a screw rod, as shown in the diagram.



DONE

2.8 Power source connection

- Before energizing, check whether the network voltage is consistent with the voltage value indicated on the equipment label.
- When connecting the equipment with the electrical system, the electrical system must be provided with a line safety device.
- When connecting the power supply line with the socket, a reliable grounding wire must be provided.
- Good cover grounding protection must be provided.
- Main power supply must be provided with an automatic leakage control opening contact, and voltage stabilizer must be provided in the area where the power supply voltage is instable.

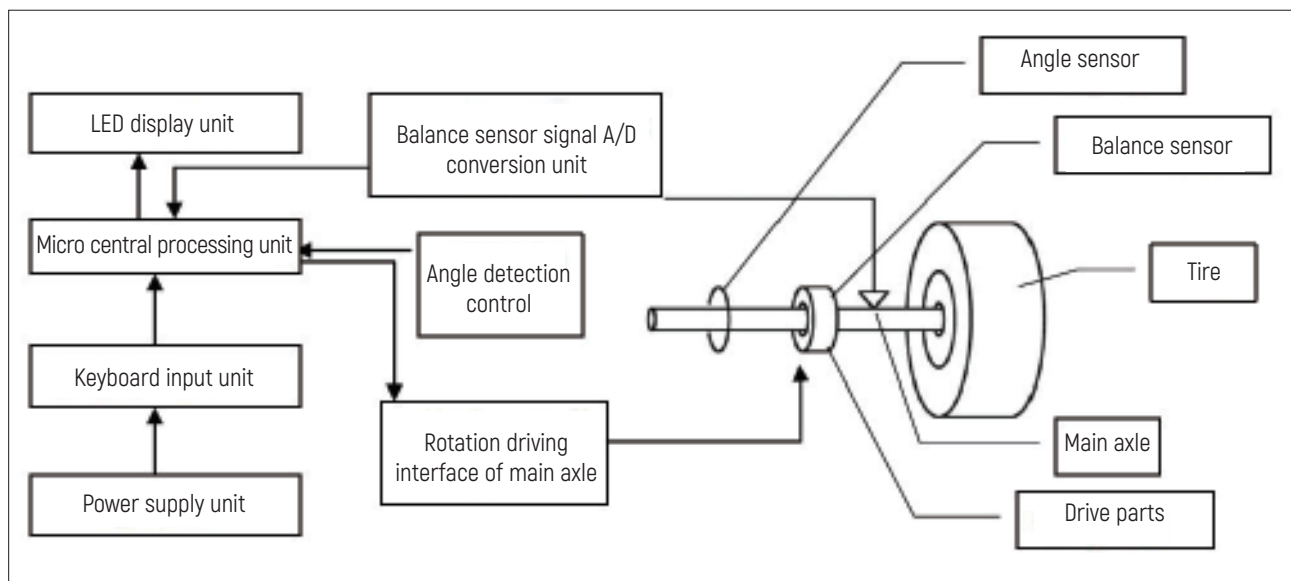
2.9 Complete machine test

- Replace the protective cover, and press "START" key. The rotation direction of transmission components of the balancing machine shall be consistent with that indicated on the label (clockwise direction). Note: After the first short rotation cycle, the data displayed on the panel can be ignored.
- In a rotation cycle, the interlocking system can be verified for normal operation by replacing or lifting up the protective cover.

Chapter III Overview of Tire Balancing Machine

Only when the micro central processor displays a "OK" message after checking that all equipment units work normally, the user can perform the balance operation.

When performing the balance operation, the micro processor controls the main axle rotation of the balancing machine via the driving interface, and the unbalance signal detected by a balance sensor is transmitted to the micro processor port via the A/D converter, so that the processor can calculate the unbalance value through overall analysis both of unbalance signal and main axle angle signal, and display it via a LED unit. Man-machine dialogue can be realized through the keyboard and LED unit.



3.2 Description of parameters

The series of balancing machine can be used for balancing of general tire and special vehicle tires (jigs must be purchased separately).

- When the manual measuring tape is drawn out along the rim and maintained on it, the distance reading (i.e., the offset measurement referred to in the Manual) shall be read, and the tire diameter on the tire label shall be checked or measured with a caliper. All measured values shall be entered into the computer. The distance value measured (offset value) provides the inner plane position of dynamic balance to the computer. The diameter value measured provides the distance from the hub center for weight application to the computer.

- When inputting the hub width into the computer, it will be added to the offset measurement automatically to determine outer plane of the hub. The hub width is input by the operator manually as W measurement.

- When the balancing machine detects any unbalance during wheel rotation, the computer will calculate the weight necessary for unbalance calibration and its application position. When the weight necessary for unbalance calibration is displayed on the control panel, the alignment light will help the operator with alignment of weight application necessary for unbalance calibration. Weight display and alignment light indicate the inner and outer plane calibration weights and positions, respectively

Chapter IV Operating Instructions

4.1 Operation interface



1. Inside unbalance value display.
2. Inside unbalance location.
3. Outside unbalance value display.
4. Outside unbalance location.
5. **【a+,a-】** Manual input the rim distance increase / decrease.
6. **【b+,b-】** Manual input rim width increase / decrease.
7. **【d+,d-】** Manual input the diameter increase / decrease.
8. **【STOP】** Stop key to stopping spin or resetting parameter.
9. **【START】** Start key to spin.
10. **【FINE】** Fine key to indicate actual unbalance weight amount reading below 5 grams or 1/4 ounce.
11. **【C】** Recalculates the unbalance weight amount to be attached on the wheel and also used to start self calibration program.

12. **【D】** Diagnostic key, press to perform the system diagnoses.
13. **【F】** F key selects static or dynamic balancing mode.
14. **【ALU】** ALU key selects ALU balancing mode.
15. **【OPT】** OPT key permits to optimize the weight to be attached to the wheel.
16. **【Gram/Ounce】** Gram / Ounce key allows user to select.
17. **【STOP+C】** To change between manual and automatic spin modes and vice –versa.
Press and hold Stop + C. All lights on display panel will turn off. This is the confirmation that spin mode had been change.
18. **【STOP+b+/b-】** Wheel width unit switching (mm/inches).
19. **【STOP+d+/d-】** Rim diameter unit switching (mm/inches).
20. **【STOP+OPT】** Rim diameter gauge calibration (only apply A and AW model).
21. **【STOP+FINE】** Wheel distance gauge calibration (only apply A and AW model).
22. **【STOP+Gram/Ounce】** Wheel width gauge calibration (only apply A and AW model).
23. **【C+F】** Self Calibration.
24. **【FINE+d+/d-】** Setting dE parameter in ALU-s mode.

Note:

1. The instant startup function of protective cover can be maintained after shutdown.
2. The function of selecting inch as the unit of rim width and rim diagram cannot be maintained after shutdown.

Note

Press the keys with hands. To press the keys with balance plier or other sharp objects is not allowed.



4.2 Tire installation

Preinstallation preparation:

- Check and remove the dust and sand from the tire, and check whether there is metal, stone and other foreign matters;
- Check whether the tire pressure complies with the value specified;
- Check whether there is any deformation of the rim alignment plane and installation holes;
- Check whether there are foreign matters, and remove the original balance lead in the rim.

Installation mode:

There are three wheel installation methods: forward alignment, backward alignment and flange alignment specifically for large and medium tire, these methods can be selected according to actual conditions.

1) Forward alignment

Forward alignment is a common alignment method, which is easy for operation and mainly applicable for common steel hub and iron hub that features minor deformation. Installation sequence:

Main → axle (rim surface is installed inwards) → cone → opening and closing nut



2) Backward alignment

In case of large outer deformation of wheel, the backward alignment can be used to ensure steel hub and main axle alignment accuracy. The alignment method is applicable to aluminum hub, especially, those thicker aluminum hubs, and the measurement precision will be higher. Installation sequence:

Main axle → suitable cone → wheel → plastic bowl → opening and closing nut



3) Alignment via large flange (optional)

The alignment method is applicable to large wheel installation. Installation sequence:

Main axle → large flange (fixed on main axle) → wheel → large cone → quick nut

Note

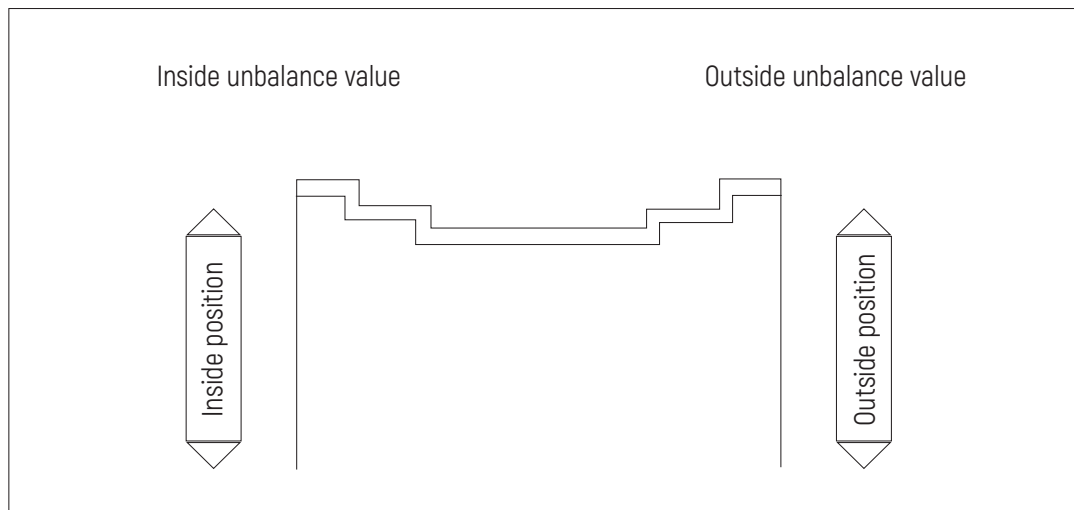
Cone selected shall be suitable to the rim center hole, and its direction; otherwise it may cause incorrect measurement.



4.3 Balance operation

Turn on the power switch, install the tire to be balanced and input the rim data. The tire starts to rotate after covering with the protective cover and pressing [START] key, and then the unbalance value will be displayed on the display after stop; the value is the weight of the balance weight to be added inside and outside, thus choose the balance weight according to the unbalance value. First, slowly turn the tire by hand until all unbalance indicators inside are on, which means that the highest point (the position of 12 o'clock) inside the rim is the unbalance position. Add the balance weight at this position, and then repeat the step mentioned above, e.g. add the corresponding balance weight outside the rim. Conduct the test again to display the tire balance [0] [0]. In actual operation, it is sometimes impossible to display [0]. There is a difference of ± 1 to 2 g, which is normal and acceptable.

Note: Generally, the balance value of tires is displayed every 5 g, while the unbalance value less than 5 g is not displayed. After each balance operation (rotation) is completed, press [FINE] key to display the actual unbalance value.



4.4 Balance mode

Need to choose different balancing modes when balance the wheel which is special and cannot add the weights at the two sides on the rim as the normal operation.

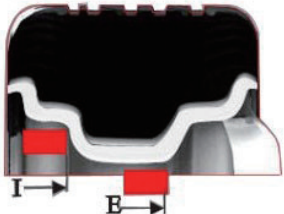
Press [ALU] to choose every dynamic balancing mode, press [F] to choose STA or DYN balancing mode.

The position to put on the weight should refer to the lamp indicator position under each balancing mode.

Normal: As the position shown in the picture, clamp the weight at two sides of the rim, suitable for steel or aluminum alloy rim.

ALU modes :

- The machine balancing program allows the user to access several wheel weight positions automatically from a spin cycle by just pressing one button. The program will automatically and without an additional spin recalculate the weight requirements and position.
- Press the —ALU || key to select desire balance mode and the corresponding indicator LED light reflecting the balancing mode will be turned on. This is a toggle switch that will move from mode to mode after every press of the key.
- Once the LED light is on the position and values of the INNER and OUTER planes are shown.

ALU —1			Balancing of light alloy rims with application of adhesive weight on rim shoulders
ALU -2			Balancing of light alloy rims with hidden application of the outer adhesive weights. Outer weight position is fixed
ALU -3			Combined application: clip on weight inside and hidden adhesive weight on outside. Outer weight position is the same as ALU 2
ALU -4			Combined application: adhesive weight outside and clip-on weight inside.
ALU -S			Measure the dimensions as shown diagram for ALU-S

Note

Some slight residual unbalances may remain at the end of the test spin due to the considerable difference in shape that found on rims with same rated dimensions. Therefore, if the standard ALU programs are not giving a satisfy balancing result, choose the variable plane program ALU-S to have correct balancing.

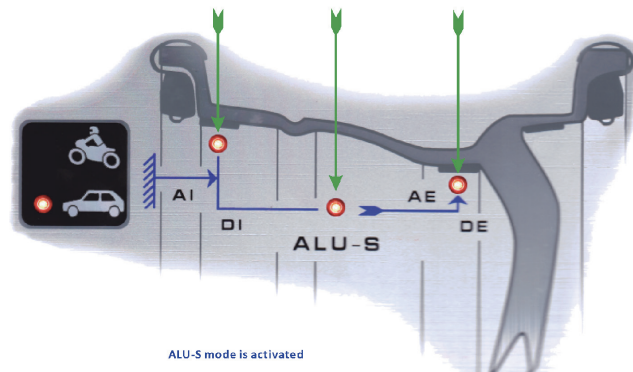


ALU-S Program

This program makes it possible to apply adhesive weights in user selected position. It is used for maximum precision balancing of light alloy rims that require both weight to be apply on the same side[INNER(left)side].

Press[ALU]key enter the program until the light over the mark "ALU-S"

illuminates, as illustrated, 3 lights on the balancing mode indicator are illuminated. Or pull out the front measuring gauge into rim two position where apply weight on A1 position and AE position. The system will automatic into ALU-S program.

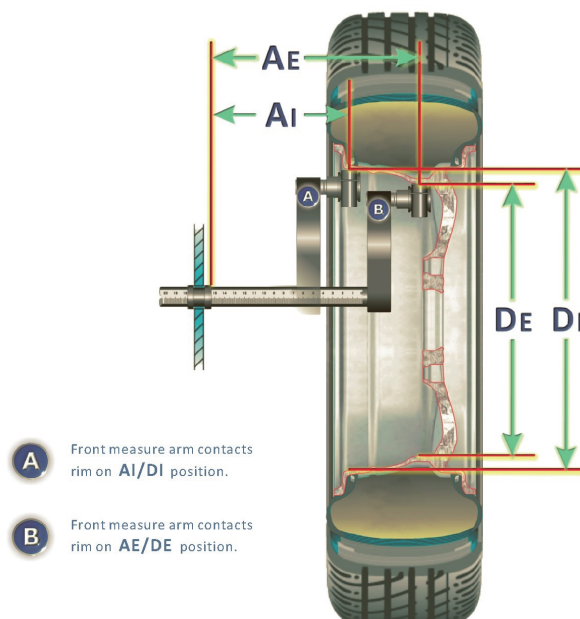


Wheel geometric data acquisition

Geometric data relating to the actuable balancing planes rather than the rated wheel data (A, W and D as in standard dynamic and standar ALU program) have to be entered. The balancing planes where the adhesive weights are to be applied may be selected by user according to the specific shape of the rim.

However, it is preferable to select balancing planes as far apart as possible in order to reduce the quantity of weights to be applied, normally, the distance between INNER(left) and OUTER(right) planes shall be more than 38mm(1.5 inches).

Definitions:	AI	=	Distance of INNER (left) plane.
	DI	=	INNER (left) plane diameter.
	AE	=	Distance of OUTER (right) plane.
	DE	=	OUTER (right) plane diameter.



Operation Steps:

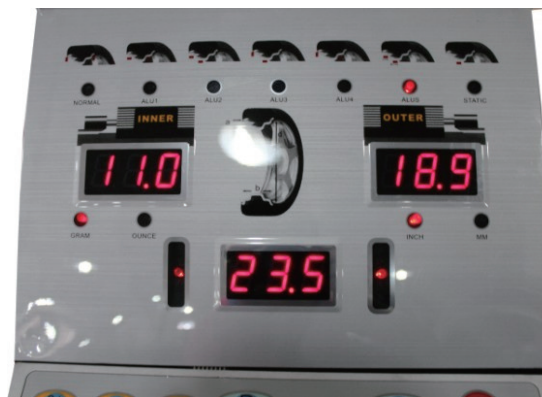
Step 1. Press key to initialize the ALU-s mode. (Note: The MACHINE will enter to ALU-s mode automatically when the measurement pointer point at 2 different position and acquired 2 position data.)

Step 2.

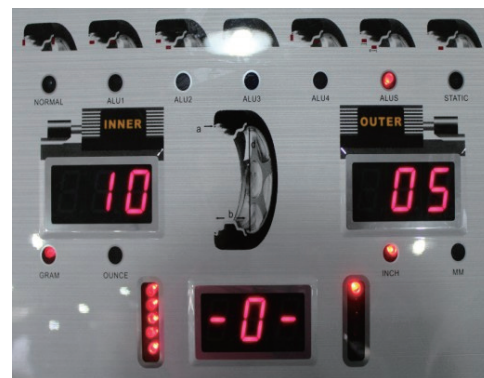
(1) Pull out the measurement pointer to 2 defined positions, during data acquiring.

(2) display panel will appear ALU-S-

(3) until the measuring pointer return back to the home position, the display will give the readings of a_l , a_E , d_l and d_E . As shown in the picture.



Step 3. Lower the protection hood and press “START” to proceed a wheel spin. The unbalance weight reading in both defined positions come out in the window, as shown in the picture.



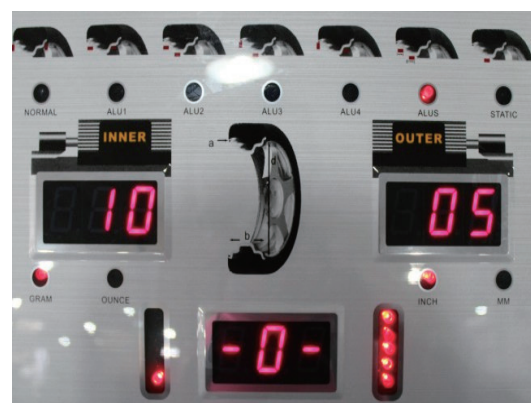
Step 4. Rotate the wheel to until the left side position indicator LED all lighted on, use foot pedal brake to hold the wheel in position.

Step 5. Pull out the pointer with need weight to find the a l position , when a l is reached, the mark [-0-] will show on the display and the buzzer is on, as shown in the picture. Attach the weight on the a l position according to the unbalance reading.



Step 6. Do the same procedure of step 4 and step5 to find the aE position. pull out the pointer with need weight to find the aE position , when dE is reached, the mark [-0-] will show on the display and the buzzer is on, as shown in the picture. Attach the weight on the aE position according to the unbalance reading.

Step 7. Spin the wheel again to find any remain.



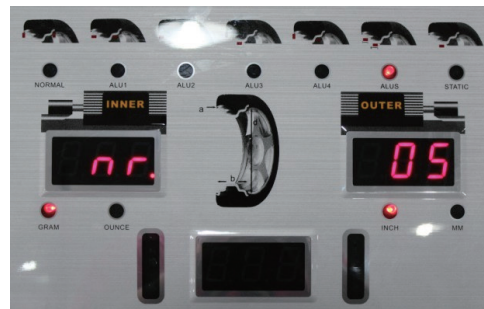
4.5 ALU-S split weight function :

Step 1. The split weight function is for attaching (hiding) adhesive weights behind wheel spokes. It is used on ALU-s mode only.

Step 2. After completed of ALU-S calculation, press D+OPT to enter split weight function. Following picture shows how many spoke the rim has.



Step 3. Press a+, b+, d+ or a-, b-, d- key to enter the number of spokes (ranging 3 to 12)

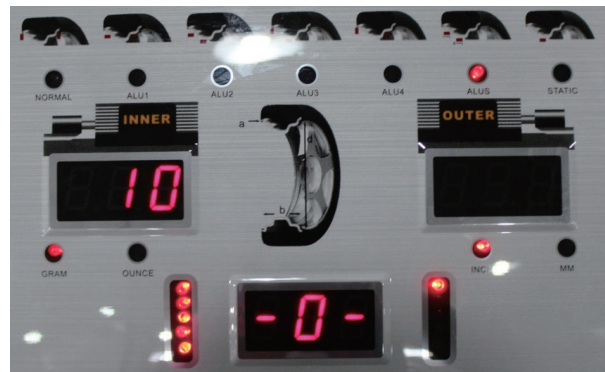


Step 4. Press D+OPT key to confirm. Then display will shows following [POS][12H].

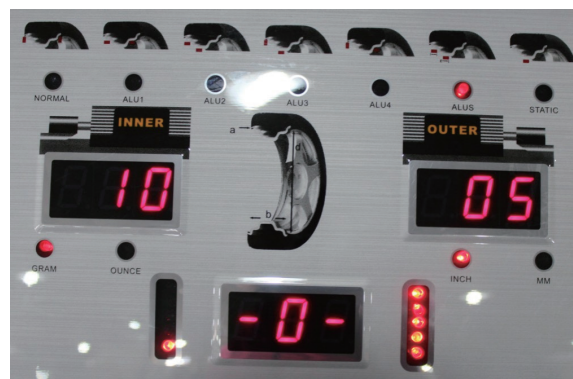


Step 5. Rotate the wheel by hand until one of spoke is at 12 o'clock position. Then press ALU key to confirm

Step 6. rotate the rim slowly until inside flash light is fully on. Place a correspond weight on distance gauge and adhesive on the rim. It should behind the spokes.



Step 7. Do the same procedure of step 6 on inner other side.



Step 8. Place a correspond weight distance gauge and adhesive on the rim. It should behind the spokes.

Step 9. ALU-S split weight function completed.

Notice: Press "C" key to release the memory. Otherwise. Automatic distance gauge didn't detect the rim size.

4.6 Optimized Balancing Function :

- This function is suggested only when STATIC balancing value is greater than 30 grams / 1 ounce to optimize the balancing and reduce total weight to be added.
- For the best balancing results follow this procedure.
- NOTE – After repositioning of tire on rim, inflate tire.

1. Press the “OPT” key and display reads as show
2. Press “START” key to spin, at wheel stop the display shows 180 at OUTER plane. The tire must be rotated on rim 180 degrees. Mark tire and rim when all right side LED lights are on. Mark cone and tire position as well. Mount tire and rim as requested by program. Inflate the tire to required pressure.
3. Remount the wheel on the balancing shaft and press the “START” key. At wheel stop the display will read out data as shown.
4. The right window indicates the static balance weight reduction rate after remounting tire in percent The left window tells the weight to be added after tire remounted. In the example, this is calculated as $100 \times (1-0.80)$ or 20 grams or 1¼ ounce.
5. complete process, rotate the wheel slowly until the display indicates tire position is reached. Mark the tire at 12 o'clock position on the tire. Again, rotate the wheel slowly until the display reads Prim. Mark the tire at 12 o'clock position on the rim. Remove the wheel from spin shaft and re-mount matching the marks on the rim and tire. One last time position wheel on spin shaft and do a normal balance. The STATIC weight requirement will be 20 grams.
6. Press “STOP” key to exit the OPT program.

Chapter V System Calibration Procedure

Note

After initial installation, maintenance and repair of the transmission shaft and piezoelectric ceramic or replacement of the circuit board, the self-calibration procedure shall be implemented to ensure the measurement accuracy of the balancing machine.

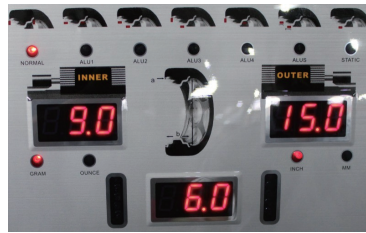
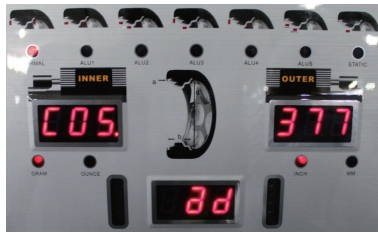


5.1 Gauge Calibration Procedures (only A and AW model)

Important procedure

- Calibration is required when the wheel balancer is first into operation at a fixed location or after is moved or when technician suspect the machine of producing incorrect data outputs.
 - Install on spin shaft a 15 || rim. The wheel {rim and tire assembly} must be in good condition with the tire fully inflated.
- An operator step by step guide follows.
- The entire sequence must be completed in order for balancer to perform accurately and fast. Failure to complete all steps

1. Turn Power on. Display shows [COS][377][ad]. Then display will turn default parameter as showing below pictures.



2. Distance gauge calibration. (only A and AW model)

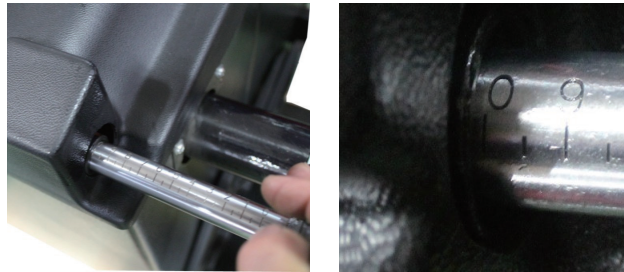
2.1 Press "STOP+FINE". Balancer start to do calibration process shown below pictures.



2.2 Find the distance gauge, located near by the spin shaft. The distance gauge had marked number.



2.3 Display shows P!0. then pull the distance gauge to find 10, and then level up with sleeve.



2.4. Hold the distance gauge and then press ALU to confirm for next process. Display now shows [-0-]



2.5 Pull the distance gauge and touch the inside of flange's surface (see below picture).



2.6 Hold the distance gauge, and then press —ALU || to confirm for next process. Display shows [CAL],[End]. Return the distance gauge to home position.

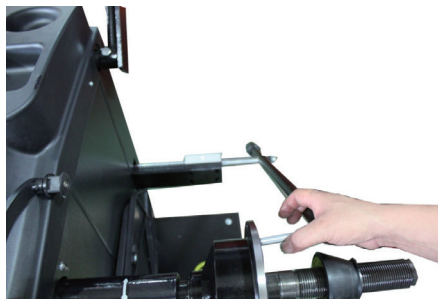


3. Width distance gauge calibration (only AW model)

3.1 Press STOP + Gram/Ounce key to enter calibration procedure shown picture below.



3.2 Display shows [CAL][OF]. Then pull the width distance gauge to touch the outside of flange's surface (see below picture).



3.3 Hold the width distance gauge and press ALU key to confirm.



4. Diameter calibration (only A and AW model)

4.1 Press STOP+OPT key to enter diameter calibration procedure.

4.2 Mount 14-16 steel rim. Press D+ or D- adjust the diameter. Then press ALU key to confirm.



4.3 Pull out the distance gauge to touch the rim edge. And then hold that position.



4.4 Press ALU to confirm to completed procedure. Then display shows [CAL][End].

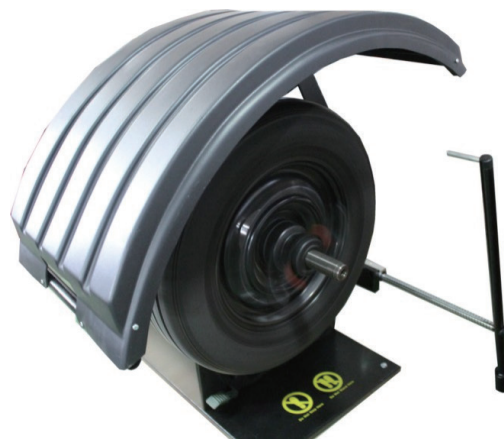


5.2 Self calibration

1. NOTE - calibration is required when the machine is first put into operation at a fixed location, after it is moved to a new location or when the technician suspects the machine of providing incorrect values.
2. Mount a good shape wheel (round and not damaged) on the spin shaft making sure the tire is fully inflated. Input the rim data parameters a, b, and d.
3. Press and hold C and F. When all the LED indicators stop flashing release buttons.



4. The machine is now in calibration mod. Press “START” key or cover the protection hood to frist spin.

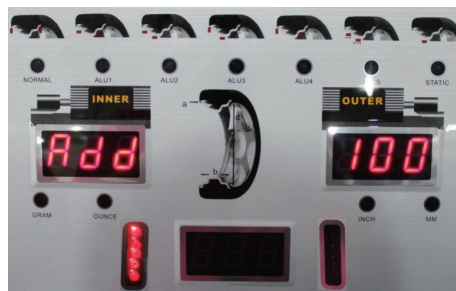


5. At first spin stop the machine will issue a display as shown. Rotate the wheel until the outer LED light fully flash. Add the 100-gram/3.5ounce wheel weight at 12 o'clock location of the wheel.



6. Press “START” key or cover the protection hood to spin to run second spin.

7. Second spin stop. The display will show following picture.



8. Rotate the wheel until the outer LED light fully flash. Add the 100-gram/3.5ounce wheel weight at 12 o'clock location of the wheel.

9. Press “START” key or cover the protection hood to spin to run third spin. At third spin top the calibration procedure is completed. Display shows following.



Error shows during balancing calibration.

[Err][-8-] forget to attach 100g or power board is out function.

[Err][-9-] forget to attach 100g

[Err][-6] sequence of outer and inner attach 100g incorrect.

[Err][-r-] hardware installation was not correct. calibration cannot completed.

Chapter VI Routine Maintenance

6.1 Disconnect the power supply before maintenance.

6.2 Adjust the belt tension

Remove the cover, loosen the motor screw, move the motor until the belt is properly tensioned, then press down the belt with force for about 4 mm lower; tighten the screws of the motor and cover it.

6.3 Replace the fuse

Two spare fuses are provided on the power panel and can be removed from the fuse holder and replaced if damaged.

6.4 Unstable balance display

A well-balanced tire is unbalanced after reinstallation, which is not a machine display error, but is generally caused by rim reinstallation. The disalignment between two installation surfaces for rim and conical surface, or the deformation of rim hole and the large gap with the taper disc as well as looseness of fast locking nut will cause large error.

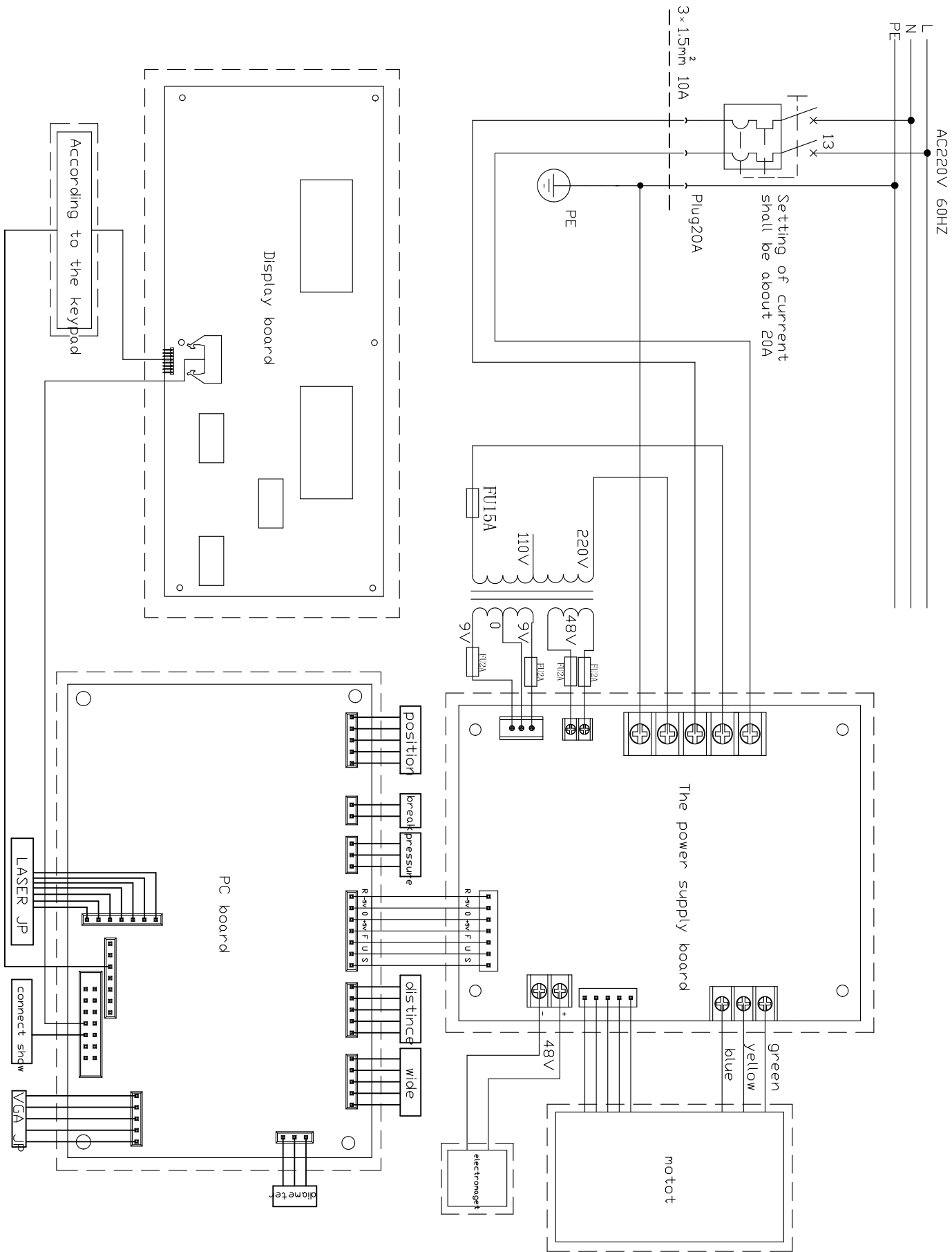
Chapter VII Common Troubleshooting Methods

Fault phenomenon	Fault causes	Troubleshooting methods
1. Turn on the power switch	1. Check the motor for damp or electric leakage	1. Dry or replace
	2. Check the power board transformer for leakage	2. Replace
	3. Check the brake resistance for leakage	3. Replace
	4. Check whether the chassis is connected to the grounding electrode line	4. Connect the grounding electrode line properly
2. No display after starting up	1. Check the external circuit for being energized	1. Use multimeter to measure
	2. Check the power switch for being damaged	2. Replace
	3. Check the fuse on the power board for being loose or burning out	3. Tighten replace it
	4. The power supply leads of computer board are broken	4. Connect the connecting wire properly
	5. Check the power panel for being damaged	5. Replace
	6. Check the computer board for being damaged	6. Replace
3. The display is normal after startup, while the motor does not rotate when pressing the start key.	9. Check the line between the main computer board and the display board for being connected properly	9. Reconnect the connecting wire properly
	1. Check the power panel for being damaged	1. Replace
	2. Check the computer board for being damaged	2. Replace
	3. Check the motor for being damaged	3. Replace
	4. Check the motor capacitance for being damaged	4. Replace
	5. Check the keyboard plate for being damaged	5. Replace
4. The motor keeps rotating after startup	6. Check the motor wire on the power board or the inner wire of the motor junction box for being loose	6. Fasten it
	1. Check the computer board for being damaged	1. Replace
	2. Check the power panel for being damaged	2. Replace
	3. Check the keyboard plate for being damaged and the position of PV board for being moved	3. Replace / adjust the position of the PV board.
5. No brake after startup	1. Check the power panel for being damaged	1. Replace
	2. Check the computer board for being damaged	2. Replace
	3. Check the brake resistance for being damaged	3. Replace
6. Display ERR1 after startup	1. Check the power panel for being damaged	1. Replace
	2. Check the computer board for being damaged	2. Replace
	3. Check the connecting wire of the position sensor for being loose or the position sensor for being damaged.	3. Re-weld or replace

Fault phenomenon	Fault causes	Troubleshooting methods
7. Display ERR2	1. Check the computer board for being damaged	1. Replace
	2. Wheels are not installed	2. Install the wheel and try again
	3. Install only one rim, but not install the tire	3. Install the tire
	4. The spindle matcher is not installed tightly	4. Re-tighten the bolt
	5. The wheel is not installed correctly and is not locked	5. Choose the proper cone and install it correctly
	6. The belt is too loose or too tight	6. Re-adjust it
	7. The screw rod is stripped and the wheel is not fastened	7. Replace it and reinstall the wheels
	8. The half nut is stripped and the wheel is not tightened.	8. Replace it and reinstall the wheels
8. Display ERR3	1. Check the computer board for being damaged	1. Replace
	2. The wheel unbalance is too serious to be calculated	2. Replace the wheel and try again or conduct the self-calibration again
9. Display ERR4	1. Check the computer board for being damaged	1. Replace
	2. If it is rotated negatively, the phase lines are connected incorrectly	2. Adjust the phase lines
	3. If it is rotated positively, there is something wrong with the photoelectric sensor	3. Adjust the position again or replace it
10. Display ERR5	1. Check the computer board for being damaged	1. Replace
	2. The protective cover of wheels is not put down	2. Put the protective cover down
	3. Check the limit switch for being damaged	3. Replace
11. Display ERR7	1. Check the computer board for being damaged	1. Replace
	2. The stored data is lost	2. Input the parameters and recalibrate it
12. Display ERR8	1. Check the computer board for being damaged	1. Replace
	2. Check the power panel for being damaged	2. Replace
	3. Do not add 100 g lead when conducting self-calibration.	3. Apply a 100g lead
	4. The lead of piezoelectric sensor is broken	4. Connect the connecting wire properly
	5. The piezoelectric sensor is broken	5. Replace
13. Display ERR78	1. The screw of width ruler potentiometer is loose and the data roller is disordered after pulling a and d	1. Re-adjust the potentiometer, tighten the screws, and carry out self-calibration of the measuring tape
14. Only 00-00 is displayed, and the value is not displayed	1. The lead of piezoelectric sensor is broken or in poor contact	1.Reconnect
	2. The stored data is lost	2.Re-correct memory value
	3. Check the computer board for damage	3. Replace

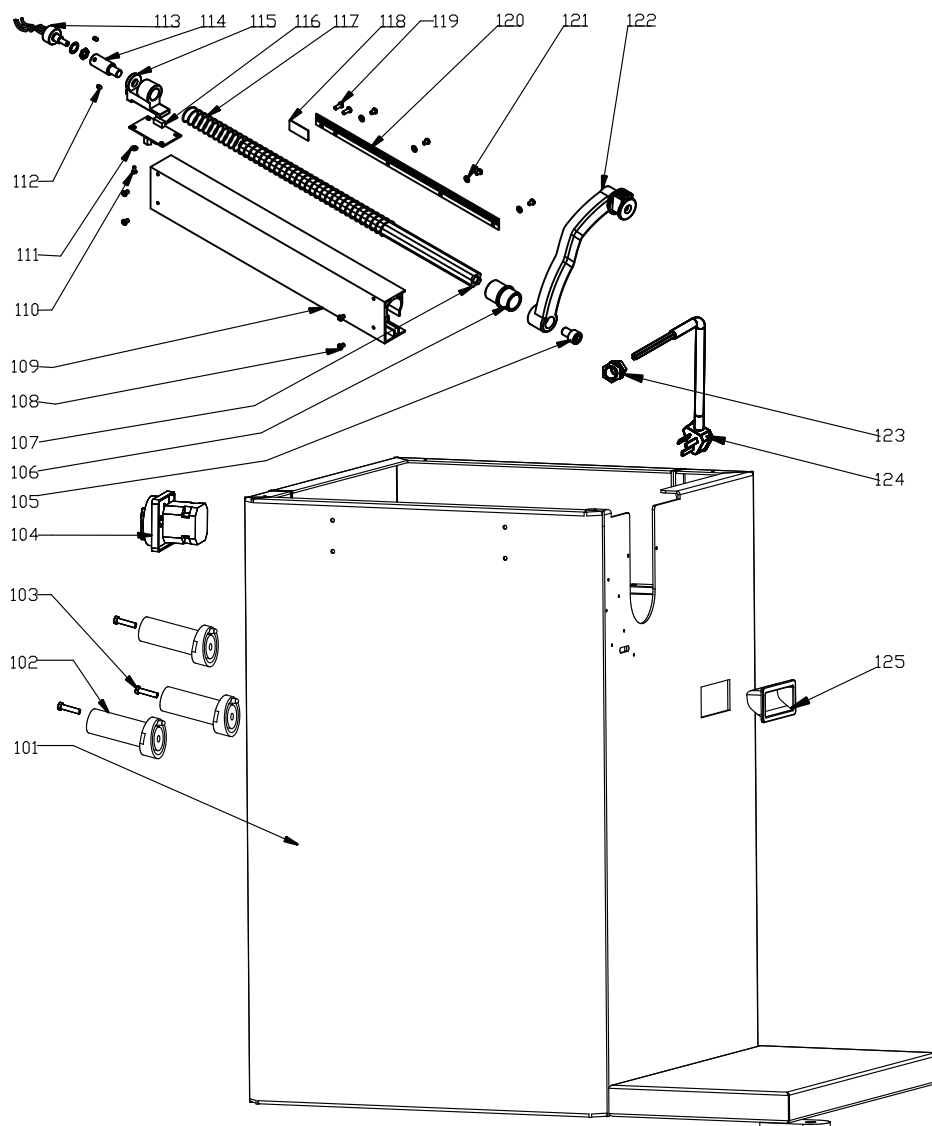
Fault phenomenon	Fault causes	Troubleshooting methods
15. The value deviates after self-calibration	1. It's possible that the wheel error is large	1. Use a standard wheel. It's best to choose a well-balanced wheel
	2. The three memory parameters are confused	2. Perform self-calibration after modifying memory parameters
16. The 100g value is not displayed correctly, it is not below	1. The wheel is not standard or there are foreign matters	1. Replace with the standard balanced wheel
	2. The 100g system is not self-calibrated properly	2. 100g system self-calibration
	3. The display value is not stable	1. Replace the wheel
17. The wheel value varies by more than 5g per rotation	1. The wheel is not standard or there are foreign matters or the installation surface of rim center is deformed	1. Replace the wheel
	2. The piezoelectric sensor is damped or the locking nut is not clamped	2. Dry and readjust the piezoelectric sensor.
	3. The external supply voltage is low or the tire pressure is insufficient; matcher	3. Stabilize and inflate
	4. The ground is uneven or the machinery is unstable	4. Put it on the horizontal floor and apply anchor screw
18. The value varies by tens of grams per rotation	1. There are foreign matters on wheel or the unbalance is large	1. Replace the wheel
	2. The piezoelectric sensor is damaged	2. Check sensors and connections
	3. The external power supply voltage is low	3. Check and repair power supply or install the voltage stabilizer

Chapter VIII Circuit Diagram



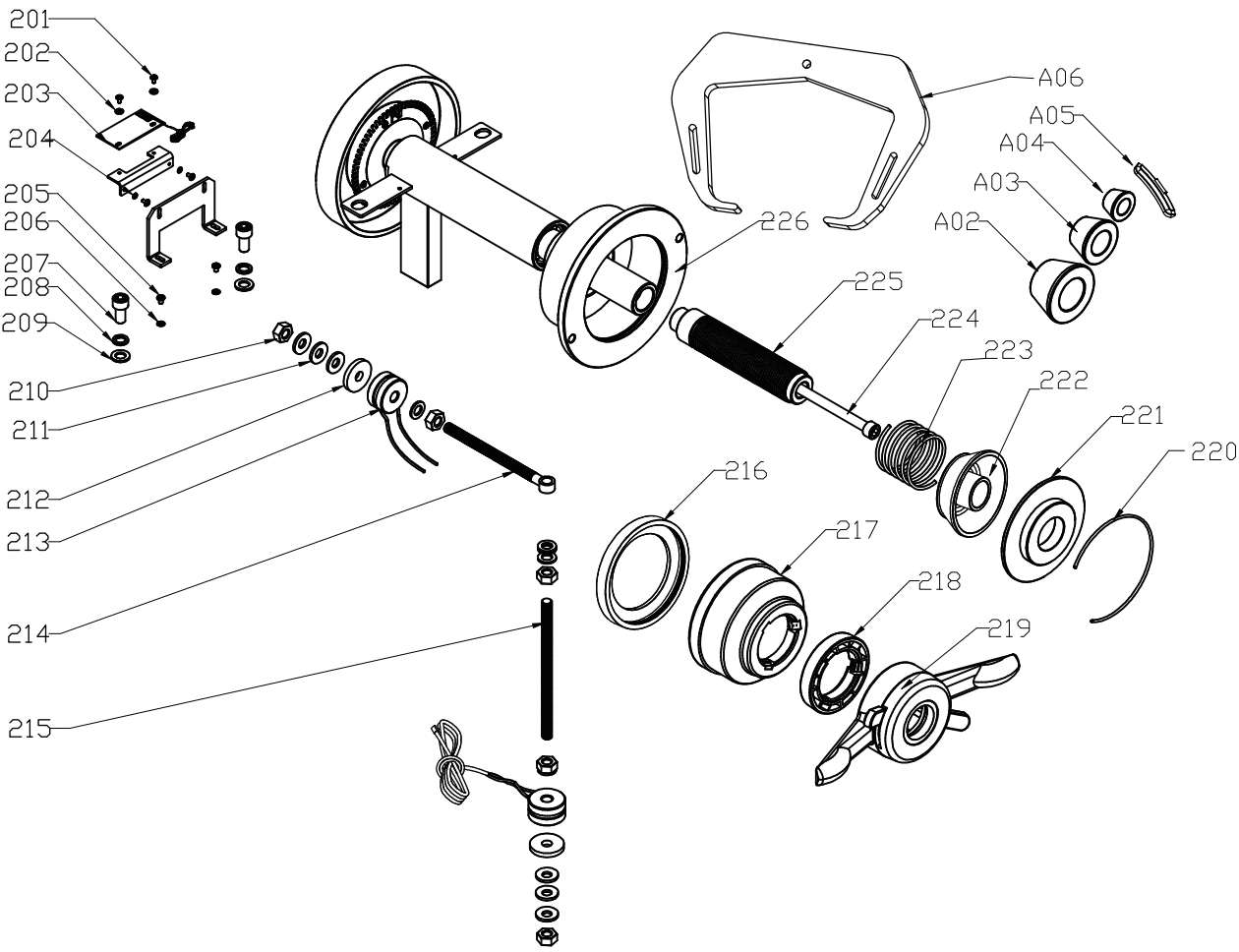
Chapter IX Product Explosion Diagram

1. Chassis:



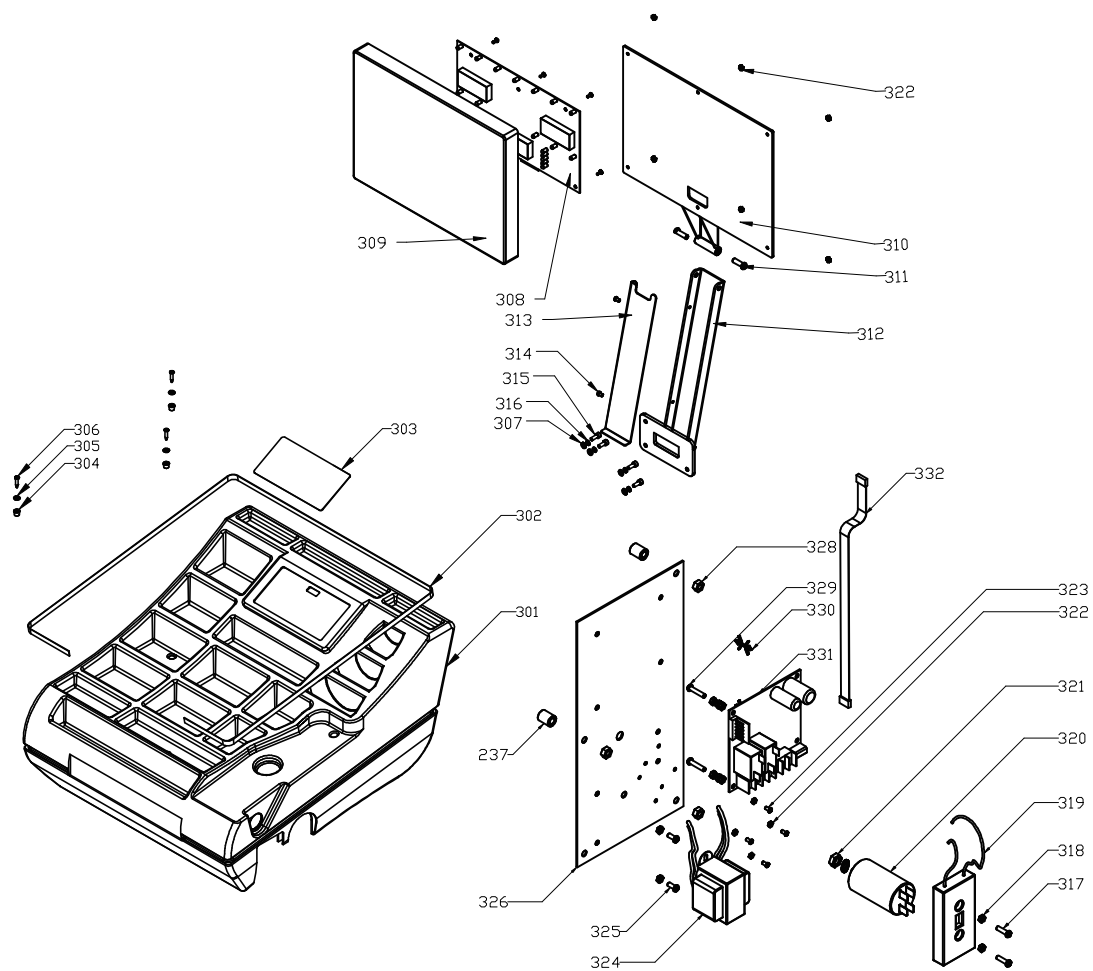
Drawing No.	Material No.	Specification name	Quantity
101	ZH20020096	Chassis	1
102	SJ11040079	Tool hanger (plastic)	3
103	LS60030038	Hexagon bolt (black) M6 * 30	3
104	KG90010005	Power switch lm12-16gly1 / 1	1
105	LS60010011	Hexagon socket head bolt M10 * 20	1
106	JG30021092	Slide sleeve of a-ruler	1
107	JG30020996	A-rule sliding rod 420mm	1
108	LS60020010	Cross recessed half round head bolt M4 * 6	8
109	JG30021000	Installation base of automatic a-protractor	1
110	LS60020024	Cross recessed half round head tapping screw 3 * 8	2
111	DQ40010031	Red steel paper flat pad m3 * 8 * 1	2
112	LS60010048	Kimi screw M4 * 6	2
113	DZ25010153-2	Potentiometer 6184-329a r1k l1.0	1
114	SJ11040221	Automatic A extension shaft	1
115	JG30020998	Potentiometer mounting block	1
116	DL15010047	Photoelectric position sensor	1
117	TH80010005	Automatic A tension spring Ø 1.2 * 22 * 520	1
118	JG30010793	Automatic A ruler device light barrie	1
119	LS60020009	Cross recessed half round head bolt M4 * 10	2
120	JG30020997	A-foot rack 330 * 16.2 (82 feet)	1
121	DQ40010032	Red steel paper flat pad M4 * 8 * 1	4
122	PJ40010074	Handle gp-puller	1
123	WJ27010012	Cable screw pg11	1
124	XC19010060	Power cord with plug 3 * 0.75 * 3M (national standard)	1
125	SJ11040260	Plastic handle (black 3.0 plate)	1

2. Transmission component device:



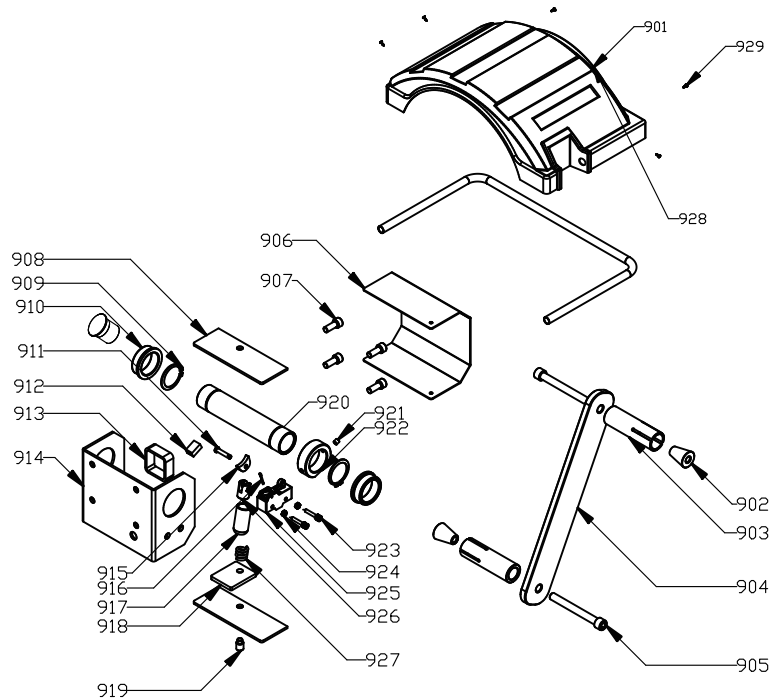
Drawing No.	Material No.	Specification name	Quantity
201	LS60020013	Cross round head gasket bolt M3 * 6	4
202	DQ40010031	Red steel paper flat washer 3*8*1	2
203	DL15010032	64 teeth photoelectric sensor + line	1
204	DQ40010026	Outer chrysanthemum gasket M3	2
205	LS60020009	Cross round head gasket bolt M4 * 10	2
206	DQ40010027	Outer chrysanthemum gasket M4	2
207	LS60010011	Hexagon cylindrical head bolt M10*20	2
208	DQ40020005	Elastic washer Ø10	2
209	DQ40010008	Flat washer Ø10*20*2	6
210	LM70010031	Reinforced nut M10	5
211	DQ40010013	Butterfly gasket Ø10*2	6
212	DQ40020013	Pressure sensor flat pad Ø10*35*4	2
213	DZ25010001	Piezoelectric sensor + wire	1
214	WJ27010046	Single-headed screw	1
215	WJ27010047	Double-headed screw	1
216	SJ11040084	Half nuts rubber pad	1
217	SJ11040083	Plastic bowl	1
218	WJ27010025	Pressure ring of half nut	1
219	SJ11040155	Half nuts Ø36	1
220	KH50010023	Steel wire clip spring (large)	1
221	SJ11040082	Large cover plate (large washer piece) Ø40	1
222	CD23010009	(Large) cone block Ø 40 (Ø100-Ø132)	1
223	TH80010032	Spring shaft flange cover Ø36mm	1
224	CD23010028	Shaft drive stud M10	1
225	CD23010031	Screw Ø36mm	1
226	CD23010053	Drive shaft assembly Ø36	1
A02	CD23010010	(Second-large) cone block Ø40 (Ø74-Ø109	1
A03	CD23010011	(Middle) cone block Ø 40 (Ø54-Ø79)	1
A04	CD23010012	(Small) cone block Ø 40 (Ø44-Ø65)	1
A05	WJ27010023	Self-calibration lead 100g	1
A06	SJ11040025	Caliper	1

3. Motor/Upper cover plate:



Drawing No.	Material No.	Specification name	Quantity
301	SJ11040245	Upper cover plate (black)	1
302	BC22050118	Balancer trim strip 12MM * 15M	2
303	DL15020039	Key board	1
304	SJ11040004-1	Environmentally-friendly foot cone 14*11*9MM	3
305	DQ40010004	Plain washer $\phi 6 \times 12 \times 1.5$	3
306	LS60020040	Cross recessed round head self-drilling bolt M4.2*25	3
307	DQ40010004	Flat washer $\phi 6 \times 12 \times 1.5$	4
308	DL15010052	Computer controlled display board	1
309	DL15020030	IMD Panel	1
310	ZH20020086	Display bracket device mounting plate assembly welds 2	1
311	LS60010113	Round head hex socket with cushion screw M8*20	2
312	ZH20020077	Display bracket assembly welds	1
313	JG30010898	Display bracket device connected to square slot cover plate	1
314	LS60020003	Cross recessed cup head bolt M5*10	2
315	LS60010023	Cylindrical head hexagon socket bolt M6*15	4
316	DQ40020003	Spring washer $\phi 6$	4
317	LS60020007	Cross semi-round head with pad bolt M4*16	2
318	LM70010022	Nut (white) M4	10
319	DZ25010006-3	Ceramic Brake resistor SQBB 100W15RJ	1
320	DZ25010208	Capacity 10UF 450VAC	1
321	LM70010019	Nut (white) M8	1
322	LM70010023	Nut (white) M3	10
323	LS60020013	Cross semi-round head with pad bolt M3*6	4
324	DZ25010014	Transformer 220V double 10V 10W	1
325	LS60020009	Cross semi-round head with pad bolt M4*10	2
326	JG30020846	Power supply aluminum plate installation	1
327	SJ11040208-1	ABS gasket (straight column) M6*15	3
328	LM70010020	Nut (white) M6	3
329	LS60020006	Cross countersunk head bolt M4*25	2
330	DZ25010004	Aircraft frame	2
331	JG30020869	Power board (220 v)	1
332	JG30020958	Power signal bar cable (5*0.3mm ²)	1

4. Protective cover:



Drawing No.	Material No.	Specification name	Quantity
901	SJ11040088	Protective cover (large SP7 series)	1
902	JG30010886	Protective cover hollow shaft locking taper sleeve	2
903	JG30010885	Protective device installed tube shaft	2
904	JG30010884	Protective cover rotating shaft connecting plate assembly	1
905	LS60010103-1	Hexagon socket bolt with round head M10 * 90	2
906	JG30010887	Protective cover outer shroud	1
907	LS60010015	Cylindrical head hexagon socket bolt M8*25	4
908	JG30010878	Protect single device mounting seat reinforcement plate	2
909	KH50010034	Circlip (for shaft) $\phi 34$	2
910	SJ11040242	Protective cover rotating shaft plastic sleeve	2
911	LS60040002	Pin shaft eye bolt 5*25	1
912	JG30010883	Protective cover device adjusts limit block	1
913	JG30010880	Protective cover device limit block	1
914	JG30010877	Protective cover mount assembly	1
915	JG30010882	Protective device to adjust the plate	1
916	LI70020025	Cotter pin M2*16	1
917	JG30010889	Protective cover adjuster connecting sleeve	1
918	JG30010879	Protective cover device adjustment plate	1
919	LS60040015	Hexagon socket countersunk jack screw M10*16	1
920	JG30010881	Protective cover mounting rotating shaft	1
921	LS60010046	Hexagon socket countersunk set screw M6*6	1
922	JG30010888	Protective cover inductive switch shaft sleeve	1
923	LS60020018	Cross recessed countersunk head bolt M4*30	2
924	LM70010022	Nut (white) M4	4
925	KG90010004-1	Microswitch Z-15GW22-B	1
926	JG30010879	Protective cover adjuster connecting shaft	1
927	TH80010054	Compression spring $\phi 16 \times 3 \times 23.5$	1
928	JG30030702	Protective cover device is installed round tube	1
929	LS60020029	Cross flat head self-drilling screws M4.8*19	5

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